

# Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: GLUE | Context: Luxembourgish Public Administration Civil Servants — GLUE Assessment

**Input Content definition:** Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

GLUE's input content is entirely English, drawn from American/British movie reviews [Q26], US/UK news [Q29, Q48], Wikipedia [Q41], Quora [Q31], linguistics journals [Q21], and English-language fiction and transcribed speech [Q38]. The dataset analysis confirms across 409 sampled examples that 'every single example reviewed across all 12 configs is in English' with US-political (Clinton, Bush, Gingrich, Ryan, Tillerson — DATASET-D10, D13, D14, D15, D22, D34), India-specific [DATASET-D16, D18], and US-pop-culture [DATASET-D3] cultural references dominating. The deployment input is trilingual Luxembourgish/French/German code-switched citizen correspondence with non-standardized Luxembourgish orthography [WEB-12, WEB-11], from a population that is 47.3% non-Luxembourgish nationals [WEB-4] with 47% of the workforce being cross-border workers from France, Belgium, and Germany [WEB-6, WEB-7]. The cultural and linguistic distance is total. IC is marked HIGH priority.

| ID     | Evidence                                                                                                                                                                                                                                                                                                            |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q26    | "The Stanford Sentiment Treebank (Socher et al., 2013) consists of sentences from movie reviews and human annotations of their sentiment." (p.3)                                                                                                                                                                    |
| Q29    | "The Microsoft Research Paraphrase Corpus (Dolan & Brockett, 2005) is a corpus of sentence pairs automatically extracted from online news sources, with human annotations for whether the sentences in the pair are semantically equivalent." (p.3)                                                                 |
| Q48    | "Examples are constructed based on news and Wikipedia text." (p.4)                                                                                                                                                                                                                                                  |
| Q41    | "The Stanford Question Answering Dataset (Rajpurkar et al. 2016) is a question-answering dataset consisting of question-paragraph pairs, where one of the sentences in the paragraph (drawn from Wikipedia) contains the answer to the corresponding question (written by an annotator)." (p.4)                     |
| Q31    | "The Quora Question Pairs dataset is a collection of question pairs from the community question-answering website Quora." (p.3)                                                                                                                                                                                     |
| Q21    | "The Corpus of Linguistic Acceptability (Warstadt et al., 2018) consists of English acceptability judgments drawn from books and journal articles on linguistic theory." (p.3)                                                                                                                                      |
| Q38    | "The premise sentences are gathered from ten different sources, including transcribed speech, fiction, and government reports." (p.4)                                                                                                                                                                               |
| WEB-12 | Neural Text Normalization for Luxembourgish — best ByT5 base achieves ~78.8% accuracy; documents 'large amount of orthographic variation' ( <a href="https://arxiv.org/html/2412.09383v1">https://arxiv.org/html/2412.09383v1</a> )                                                                                 |
| WEB-11 | LuxBorrow corpus with token-level LID and code-switching annotation across LU/DE/FR/EN — confirms code-switching as systematic input characteristic ( <a href="https://arxiv.org/html/2603.10789">https://arxiv.org/html/2603.10789</a> )                                                                           |
| WEB-4  | 47.3% foreign nationals as of 1 January 2024; over 170 nationalities — confirms input population's linguistic diversity ( <a href="https://luxembourg.public.lu/en/society-and-culture/population/demographics.html">https://luxembourg.public.lu/en/society-and-culture/population/demographics.html</a> )         |
| WEB-6  | ~47% of workforce are frontaliers — establishes the frontier-vs-resident label requirement ( <a href="https://statistiques.public.lu/en/publications/series/regards/2025/regards-01-25.html">https://statistiques.public.lu/en/publications/series/regards/2025/regards-01-25.html</a> )                            |
| WEB-7  | <a href="https://www.oecd.org/en/publications/2025/04/oecd-economic-surveys-luxembourg-2025_3eb782b5/full-report/reviving-productivity-growth_4509ab88.html">https://www.oecd.org/en/publications/2025/04/oecd-economic-surveys-luxembourg-2025_3eb782b5/full-report/reviving-productivity-growth_4509ab88.html</a> |
| D10    | I did not mention Monica in my lecture, but the first question I was asked was how President Clinton could do his job with all the distractions caused by the Monica Lewinsky affair. — US political context (Clinton/Lewinsky); culturally US-specific                                                             |
| D16    | What is black money and how can it effect the economy of a country? / "What is the effect of black money on India's macro economy? — Quora question pair; Indian economic/policy context                                                                                                                            |
| D3     | the weird thing about the santa clause 2 , purportedly a children 's movie , is that there is nothing in it to engage children emotionally . — Full sentence from American movie review; US pop-culture reference                                                                                                   |

|            |                                                                                                                                                                                                                                                                        |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>D13</b> | Platinum prices soared to 23-year highs earlier this year after President Bush (news - web sites) proposed investing \$ 1.2 billion for research on fuel cell-powered vehicles. — US political and financial news paraphrase                                           |
| <b>D14</b> | Mr. Bush had sought to store his papers in his father's presidential library, where they would have stayed secret for a half-century. — US presidential politics; US-specific cultural/political context                                                               |
| <b>D15</b> | Who succeeded Newt Gingrich as Speaker?" / "In 1998, with Speaker Newt Gingrich announcing his resignation... — US political history QA; requires US-specific knowledge                                                                                                |
| <b>D22</b> | House Speaker Paul Ryan was facing problems uniquely from fellow Republicans dissatisfied with his leadership." / "House Speaker Paul Ryan was facing problems uniquely from fellow Republicans supportive of his leadership. — US political content in diagnostic set |
| <b>D34</b> | The announcement of Tillerson's departure sent shock waves across the globe." / "People across the globe were prepared for Tillerson's departure. — US political news (Rex Tillerson/Trump administration) in diagnostic set                                           |
| <b>D18</b> | Are the notes of Rs. 2000 really embedded with a GPS chip?" / "How does the embedded NGC technology of the Rs. 2000 note works? — India-specific currency question; non-EU political context                                                                           |

## Strengths

- MNLI's deliberate genre diversity — ten sources including government reports, transcribed speech, and fiction [Q38] — at least demonstrates a methodological awareness that NLU evaluation should span register variation, a principle transferable to a custom Luxembourgish suite even though GLUE itself does not realize it for the target language.
- The diagnostic set draws on naturally-occurring sentences from multiple domains [Q63], offering a genre-diversity template for a custom Luxembourgish administrative diagnostic.

| ID         | Evidence                                                                                                                                                                                                                               |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q38</b> | "The premise sentences are gathered from ten different sources, including transcribed speech, fiction, and government reports." (p.4)                                                                                                  |
| <b>Q63</b> | "We ensure the data is reasonably diverse by producing examples for a variety of linguistic phenomena and basing our examples on naturally-occurring sentences from several domains (news, Reddit, Wikipedia, academic papers)." (p.5) |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                                        |
|-------------|----------------------------------------------------------------------------|
| <b>IC-1</b> | Do inputs require region-specific cultural/geographic/dialectal knowledge? |
| <b>IC-2</b> | Does culturally sensitive content align with target culture?               |
| <b>IC-3</b> | Flag inputs assuming Western-specific knowledge                            |
| <b>IC-4</b> | Regional annotator recruitment for culturally sensitive instances          |
| <b>IC-5</b> | Document content issues harming content validity                           |

### Checklist responses:

**IC-1:** Yes — the deployment requires region-specific knowledge including Luxembourgish lexicon and orthographic variation [WEB-12, WEB-15], French/German/Luxembourgish code-switching patterns [WEB-11], frontier-vs-resident terminology, EU institutional vocabulary, communal-vs-État administrative terminology, and continental-European formal correspondence conventions. None of these are present in GLUE [Q21, Q26, Q29, Q31, Q35, Q41].

**IC-2:** Cultural alignment is absent. The dominant cultural content is American (US politics, US sports, US movies, US-centric Quora threads — DATASET-D10, D13, D14, D17, D20, D22, D34, D42) with some Indian content [DATASET-D16, D18]. No Luxembourgish or even Luxembourg-adjacent (Belgian/French frontier) administrative content appears in the 409-example sample.

**IC-3:** Yes — Western-specific knowledge that does not transfer is pervasive: US presidential history [DATASET-D10, D14], US congressional politics [DATASET-D15, D22], American film criticism conventions [DATASET-D1, D2, D3, D4, D29, D30], antebellum American fiction dialect [DATASET-D11, D28], American telephone-speech register [DATASET-D33], and

English-specific morphosyntactic violations [DATASET-D8, D40].

**IC-4: INSUFFICIENT DOCUMENTATION** — GLUE does not document recruitment of regional annotators, and no Luxembourgish annotator pool exists in the inherited datasets. The YAML notes that no Luxembourgish administrative-correspondence corpus is publicly available and CTIE access would be required.

**IC-5:** The content-validity violation is severe: the input distribution shares neither language, script-norm domain, register, nor cultural reference frame with the deployment. Construct-irrelevant variance from US/Indian cultural specificity will dominate any signal a model carries from GLUE.

| ID     | Evidence                                                                                                                                                                                                                                                                                                                                                                                         |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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| WEB-12 | Neural Text Normalization for Luxembourgish — best ByT5 base achieves ~78.8% accuracy; documents 'large amount of orthographic variation' ( <a href="https://arxiv.org/html/2412.09383v1">https://arxiv.org/html/2412.09383v1</a> )                                                                                                                                                              |
| WEB-15 | spellux Python package documents homograph density and orthographic variation as ongoing Luxembourgish NLP challenges ( <a href="https://github.com/questoph/spellux">https://github.com/questoph/spellux</a> )                                                                                                                                                                                  |
| Q21    | "The Corpus of Linguistic Acceptability (Warstadt et al., 2018) consists of English acceptability judgments drawn from books and journal articles on linguistic theory." (p.3)                                                                                                                                                                                                                   |
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| Q31    | "The Quora Question Pairs dataset is a collection of question pairs from the community question-answering website Quora." (p.3)                                                                                                                                                                                                                                                                  |
| Q35    | "The Semantic Textual Similarity Benchmark (Cer et al., 2017) is a collection of sentence pairs drawn from news headlines, video and image captions, and natural language inference data. Each pair is human-annotated with a similarity score from 1 to 5; the task is to predict these scores. Follow common practice, we evaluate using Pearson and Spearman correlation coefficients." (p.3) |
| Q41    | "The Stanford Question Answering Dataset (Rajpurkar et al. 2016) is a question-answering dataset consisting of question-paragraph pairs, where one of the sentences in the paragraph (drawn from Wikipedia) contains the answer to the corresponding question (written by an annotator)." (p.4)                                                                                                  |
| D10    | I did not mention Monica in my lecture, but the first question I was asked was how President Clinton could do his job with all the distractions caused by the Monica Lewinsky affair. — US political context (Clinton/Lewinsky); culturally US-specific                                                                                                                                          |
| D16    | What is black money and how can it effect the economy of a country?" / "What is the effect of black money on India's macro economy? — Quora question pair; Indian economic/policy context                                                                                                                                                                                                        |
| D13    | Platinum prices soared to 23-year highs earlier this year after President Bush (news - web sites) proposed investing \$ 1.2 billion for research on fuel cell-powered vehicles. — US political and financial news paraphrase                                                                                                                                                                     |
| D14    | Mr. Bush had sought to store his papers in his father's presidential library, where they would have stayed secret for a half-century. — US presidential politics; US-specific cultural/political context                                                                                                                                                                                         |
| D17    | How do mountain ranges form, and what are some of the major mountain ranges in Oklahoma?" / "How do mountain ranges in Oklahoma differ from mountain ranges in Idaho? — US geography; Quora community context                                                                                                                                                                                    |
| D20    | In-form Rooney's hot goalscoring streak of seven goals in his last four internationals saw him win the vote to be crowned England's Player of the Year for 2008. — British sports news; no Luxembourg administrative relevance                                                                                                                                                                   |
| D22    | House Speaker Paul Ryan was facing problems uniquely from fellow Republicans dissatisfied with his leadership." / "House Speaker Paul Ryan was facing problems uniquely from fellow Republicans supportive of his leadership. — US political content in diagnostic set                                                                                                                           |
| D34    | The announcement of Tillerson's departure sent shock waves across the globe." / "People across the globe were prepared for Tillerson's departure. — US political news (Rex Tillerson/Trump administration) in diagnostic set                                                                                                                                                                     |
| D42    | Grisham did not win the popular vote." / "Grisham almost won the popular vote. — US electoral context in diagnostic set                                                                                                                                                                                                                                                                          |
| D18    | Are the notes of Rs. 2000 really embedded with a GPS chip?" / "How does the embedded NGC technology of the Rs. 2000 note works? — India-specific currency question; non-EU political context                                                                                                                                                                                                     |
| D15    | Who succeeded Newt Gingrich as Speaker?" / "In 1998, with Speaker Newt Gingrich announcing his resignation... — US political history QA; requires US-specific knowledge                                                                                                                                                                                                                          |
| D1     | sleepwalk through vulgarities in a sequel you can refuse — English movie-review fragment; sentiment label calibrated to film criticism register                                                                                                                                                                                                                                                  |
| D11    | Yes, Mistuh Reese, suh? — Dialect representation of African American vernacular in fiction                                                                                                                                                                                                                                                                                                       |

|       |                                                                                                                                                                                                                                                                                                             |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D33   | oh yeah no that's uh that's a that's a real interesting movie and it's got a good historical perspective to it — Transcribed telephone speech; casual American spoken English                                                                                                                               |
| D8    | They drank the pub. — Idiomatic English usage; grammatical acceptability judgment                                                                                                                                                                                                                           |
| D2    | at its best moments — Sub-sentential movie-review fragment; context-free without film domain                                                                                                                                                                                                                |
| D3    | the weird thing about the santa clause 2 , purportedly a children 's movie , is that there is nothing in it to engage children emotionally . — Full sentence from American movie review; US pop-culture reference                                                                                           |
| D4    | no lika da — Non-standard English fragment (phonetic representation); unclear sentiment signal                                                                                                                                                                                                              |
| D29   | the horrors — Two-word fragment; highly context-dependent sentiment label                                                                                                                                                                                                                                   |
| D30   | heroes — Single-word fragment; no sentential context                                                                                                                                                                                                                                                        |
| D28   | Yes, Mistuh Reese, suh?" / "The slave spoke to Mr Reece. — Dialect transcription from American fiction (antebellum setting)                                                                                                                                                                                 |
| D40   | The children eat all chocolate. — English determiner usage; grammaticality judgment                                                                                                                                                                                                                         |
| WEB-4 | 47.3% foreign nationals as of 1 January 2024; over 170 nationalities — confirms input population's linguistic diversity ( <a href="https://luxembourg.public.lu/en/society-and-culture/population/demographics.html">https://luxembourg.public.lu/en/society-and-culture/population/demographics.html</a> ) |
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### Information Gaps

- No annotator demographic information is published for the inherited GLUE component datasets, so the regional-annotator question cannot be addressed from documentation.

### Requires Expert Verification

- Sub-national linguistic variation within Luxembourg (Minett, Ardennes, Moselle, Stad Lëtzebuerg) requires sociolinguistic expert elicitation beyond available web sources.

### Remediation

**Gap 1: GLUE input content is entirely English (US/UK news, movie reviews, Wikipedia, Quora) with no Luxembourgish, French, or German text and no administrative-domain content.**

**Recommendation:** Build a Luxembourgish administrative-correspondence corpus from MyGuichet.lu data (with appropriate CTIE access and CNPD/GDPR review), supplementing with LuxBorrow [WEB-11] for code-switching patterns and adapting ItzGLUE's content domains. Ensure the corpus reflects the input population's linguistic diversity (47.3% non-Luxembourgish nationals, ~47% frontier workforce).

| ID     | Evidence                                                                                                                                                                                                                                  |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WEB-11 | LuxBorrow corpus with token-level LID and code-switching annotation across LU/DE/FR/EN — confirms code-switching as systematic input characteristic ( <a href="https://arxiv.org/html/2603.10789">https://arxiv.org/html/2603.10789</a> ) |